MIL-E-22437E(SH)
2 November 1981
SUPERSEDING
MIL-E-22437D(SHIPS)
24 March 1975
(See 6.5)

MILITARY SPECIFICATION

EQUIPMENT, DEGAUSSING, FOR MCM SHIPS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, construction, and testing of degaussing equipment to be used on nonmagnetic mine countermeasure ships. The equipment is a component of the ship's degaussing system, and will supply regulated currents to the ship's degaussing coils to compensate for the ship's permanent and induced magnetism for all conditions of the ship's heading, roll, pitch, and geographical location.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1075

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

MILITARY (Continued)

MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

MIL-STD-454 - Standard General Requirements for Electronic Equipment.

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-721 - Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety.

MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Critieria of Shipboard Equipment.

MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.

DOD-STD-1399, Section 300 - Electric Power, Alternating Current (Metric).

MIL-STD-1629 - Procedures for Performing a Failure Mode and Effect Analysis for Shipboard Equipment.

HANDBOOK

MILITARY

MIL-HDBK-472 - Maintainability Prediction.

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1.1), the contractor shall furnish unit(s) for first article inspection and approval (see 4.6 and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

3.2 Definitions. The definitions of terms used in this specification shall be in accordance with MIL-E-16400 and MIL-STD-721.

3.3 Materials. The selection and application of materials shall be in accordance with MIL-E-16400 and the additional requirements specified in 3.4.8.

3.3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incororated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Design/construction.

3.4.1 The degaussing equipment described herein is to be used on nonmagnetic mine countermeasures ships in a degaussing system comprised of:

(a) Degaussing equipment (see 3.4.5).
(b) Equipment manual (see 3.20).
(c) Degaussing coil installation (see 3.4.3).
(d) Remote alarm (see 3.4.5.11.2).
(e) Ship's degaussing folder.

3.4.1.1 The degaussing equipment type designation (example: type EMS-9A) will be furnished by the Naval Sea Systems Command (NAVSEA) after the award of the contract. The contractor shall request type designation for inclusion in the technical manual and on identification plates.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

3.4.2 The equipment shall provide a three-axes magnetometer for monitoring and measuring in microtesla (μT), the components of the earth's magnetic field parallel to the ship's vertical, athwartship, and longitudinal axes. Measured field components shall be functions of the components of the earth's magnetic field and the heading, roll, pitch of the ship, as specified in the following formulae:

$$\emptyset_M = H \cos a \sin p \cos r - H \sin a \sin r + Z \cos r \cos p$$

$$\emptyset_A = H \sin a \cos r + H \cos a \sin r \sin p + Z \sin r \cos p$$

$$\emptyset_L = H \cos a \cos p - Z \sin p$$

Where:

H = horizontal component of the earth's magnetic field in μT (always considered positive with above formulae)

Z = vertical component of earth's magnetic field in μT (positive in the northern magnetic hemisphere)

a = angle of heading of the ship relative to magnetic north measure clockwise from magnetic north to bow.

r = angle of roll about the longitudinal axis of the ship (positive for top of vertical axis to move port).

p = angle of pitch about a horizontal line perpendicular to the fore-aft axis of the ship (positive for bow to move upward).

$\emptyset_M$, $\emptyset_A$, and $\emptyset_L$ = the earth's magnetic field components parallel to the ship's vertical, athwartships and longitudinal axes, respectively.

3.4.3 The equipment shall furnish controlled currents to the loops in the M, A, and L degaussing coils on the ship. The equipment shall, when set up for automatic operation, automatically regulate the degaussing loop currents in a manner that will provide compensation for the ship's permanent and induced magnetism, regardless of the ship's heading, roll, pitch, or geographical location. In the automatic mode of operation, the M, A, and L loop currents shall be controlled by signals from the three-axes magnetometer. These currents shall be functions of the components of the earth's magnetic field and the heading, roll, and pitch of the ship, as specified in the following formulae:

$$I_M = F(\emptyset_M, K_I, F_{T1}, F_{T2}) \pm I_p$$

$$I_A = F(\emptyset_A, K_I, F_{T1}, F_{T2}) \pm I_p$$

$$I_L = F(\emptyset_L, K_I, F_{T1}, F_{T2}) \pm I_p$$

Where:

I_M, I_A, I_L = degaussing loop current in amperes for the M, A, and L coils, respectively.

F = transfer function for induced current component (see 3.4.5.9).

K_I, F_{T1}, and F_{T2} are operator settable gains or transfer functions (see 3.4.5.9).

I_p = permanent current component. Numerical value of I_p will range between zero and the maximum specified (see 3.4.5.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

3.4.3.1 A means for manually operating one or more or any combination of the M, A, or L axes in the event of failure of the magnetometer control shall be provided. The system shall be such that an operator is not required to continually update the manual course correction control during normal magnetometer control.

3.4.4 The equipment shall be operated by turning it on and periodically (daily, weekly) checking its operation. Operating personnel will not stand by during operation. When a fault occurs, trouble indicators shall sound the alarm and indicate the location of the fault. Maintenance technicians should be able to quickly isolate the fault and repair the equipment by the replacement of a modular assembly.

3.4.4.1 Power source. The equipment shall operate from a type I, 440 volt root mean square (Vrms), 60 Hertz (Hz), three-phase, standard alternating current (a.c.) shipboard power system in accordance with DOD-STD-1399, section 300. The equipment shall be capable of withstanding, without damage, the specified transients and power interruptions. The equipment shall operate as a balanced three-phase load on the power source. Criterion for balanced load shall be as defined in DOD-STD-1399, section 300. The equipment shall be connected by a three-wire, three-phase connection, and shall not constitute a ground or require grounding, either directly or by means of filter capacitors, of the ship's service power source.

3.4.4.1.1 A 115V, 60 Hz source may be used for anticondensation heaters (see 3.4.5.1.2).

3.4.5 Degaussing equipment. Degaussing equipment shall be comprised of the following items:

(a) One master cubicle (see 3.4.5.1).
(b) Three power supply cubicles (see 3.4.5.2).
(c) One magnetometer head assembly (see 3.4.5.3).
(d) One alinement fixture (see 3.4.5.4).
(e) One set of alinement tools (see 3.4.5.5).
(f) One bridge control unit (see 3.4.5.6).

3.4.5.1 Master cubicle. The master cubicle shall contain all devices and features required to make the equipment complete, except those included in the magnetometer head assembly, the power supply cubicles, or the bridge control unit. The cubicle shall contain all of the controls associated with the magnetometer, trouble indicators, centralized monitoring and control, and the backup circuits. This cubicle shall contain the control assembly (see 3.4.5.7), and 18 identical power supplies for providing currents to 18 degaussing loops (see 3.4.5.9).

3.4.5.1.1 The master cubicle shall be a dripproof assembly suitable for deck mounting. The cubicle outline shall be within the following dimensions:

Height	72 inches
Width	24 inches
Depth	26.75 inches

The weight shall be less than 550 pounds (lbs).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

3.4.5.1.2 The master cubicle shall include a lockable door assembly. Three lights shall be visible to the operator without opening the door. Lights shall be as follows:

(a) Anticondensation heaters ON.
(b) Degaussing system working to the required performance.
(c) Degaussing system fault.

All other indicators, controls, or devices shall be behind the door.

3.4.5.1.3 The master cubicle shall receive power from the ship's service power distribution system and distribute power to all other units in the system. The master cubicle shall contain the main power ON/OFF switch which will also supply power to the power supply cubicles. With the switch in the OFF position, anticondensation heaters will automatically be on. An hour meter shall be included to measure the total number of hours that the master cubicle has been ON.

3.4.5.2 Power supply cubicles. Each power supply cubicle shall contain 24 identical power supplies (see 3.4.5.9) for providing current to 24 degaussing loops. Each power supply cubicle shall:

(a) Receive power and magnetometer output signals from the master cubicle.
(b) Interface with the centralized monitoring/fault indication network in the master cubicle.
(c) Provide current to degaussing loops in any of the three axes (that is, M, A, or L axes).
(d) Be designed such that it can be located remote from the master cubicle anywhere on the ship.
(e) Be a dripproof deck mounted assembly, with the same size and mounting as the master cubicle (see 3.4.5.1.1).
(f) Have a lockable door assembly.

3.4.5.2.1 Each power supply cubicle shall have three indicator lights visible to the operator without opening the door. These indicators shall be a power ON indicator, a cubicle fault indicator, and a heater ON indicator. All other fault indicators, all current controls, and all other indicators, adjustments, or switches shall be behind the door.

3.4.5.3 Magnetometer head assembly. The magnetometer head assembly shall consist of three second harmonic type fluxgates orthogonally mounted in an enclosure which shall withstand deterioration from weather in an exposed, unsheltered, location on the ship. The enclosure shall contain all the accessories and features required for mounting on the alinement fixture (see 3.4.5.4).

3.4.5.3.1 The magnetometer head assembly shall be a passive device consisting of the fluxgates and resistors, if necessary, for consistent performance. Other electronic parts, such as transistors, capacitors, diodes, and the like, that would reduce the reliability of the fluxgates shall not be used in the magnetometer head assembly. The magnetometer head assembly shall be a replaceable, plug-in, subassembly. Spare head assemblies, supplied as repair parts (see 6.3), shall be interchangeable without affecting the mechanical or electrical alinement of the equipment, except the electrical gain of the magnetometer (see 3.4.5.7).

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

3.4.5.3.2 The magnetometer head assembly fluxgates shall be magnetically alined within plus or minus 6 minutes of arc with the assembly's reference faces.

3.4.5.4 Alinement fixture. An alinement fixture shall be provided for mounting and alining the magnetometer head assembly on the ship. The alinement fixture shall allow any alinement given sight of an azimuth bench mark and access to the ship's leveling plate. The magnetometer head assembly, when mounted on the alinement fixture, shall be capable of being lined up in azimuth and parallel with the ship's leveling plate. Alinement of the magnetometer head assembly reference faces shall be within plus or minus 6 minutes of arc. Construction of the alinement fixture shall be such that a magnetometer head assembly can be replaced without the fixture being realined.

3.4.5.5 Alinement tools. Alinement tools shall be used only to aline the magnetometer head assembly with the ship's reference plate and bench mark. They are not used during replacement of the magnetometer head or routine maintenance of the equipment.

3.4.5.5.1 The contractor shall supply the number of sets of alinement tools as specified in the contract (see 6.1.1). The procedure for alinement and the alinement tools shall be completely described in the equipment manual (see 3.20).

3.4.5.6 Bridge control unit. This unit shall be mounted in the pilot house to provide degaussing status information and a means for manual control of power supply signals, in the event of a failure in the magnetometer circuits.

3.4.5.6.1 The manual control shall be designed such that the manually controlled currents are always within 11 percent of the ideal currents for the ship's heading and location. Manual control shall be obtained with three auto-manual switches (one for each axis): one northern/southern hemisphere switch (for M axis) and one heading switch (for A and L axes). The manual control shall be designed such that an axis can be switched from automatic to manual in a mine field without making any adjustments or setting any switches, except those mentioned herein.

3.4.5.6.2 The bridge control unit shall be designed for dark adapted areas in accordance with MIL-E-16400.

3.4.5.6.3 The bridge control unit indicators shall indicate degaussing systems status, auto or manual control, and system faults. Fault indicators shall indicate fault location and when the system or an axis should be switched to manual control.

3.4.5.7 Control assembly. The control assembly shall be located in the master cubicle and shall contain the magnetometer circuits, the test/calibration circuits, and the fault isolation circuits.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T12:19Z
Check the source to verify that this is the current version before use.

3.4.5.8 Magnetometer performance. The magnetometer shall comply with the following performance requirements:

(a) The magnetic field measurement range shall be 0 to plus or minus 60 μT.
(b) Sensitivity (gain) shall be 0.100 volts per microtesla (V/μT), plus or minus 0.50 millivolts per microtesla (mV/μT) (that is, 6V out with a 60 μT field).
(c) Linearity shall be such that when the applied field is varied from plus 60 μT to minus 60 μT, the output voltage shall not vary more than 10 millivolts (mV) from a straight line defined by the actual data points taken at plus 60 μT and minus 60 μT.
(d) Maximum offset voltage under any conditions for zero magnetic field shall be no more than 25.0 mV.
(e) The zero shift due to temperature changes shall be less than 20 mV when the temperature of the magnetometer head assembly and the master cubicle are varied over their specified temperature ranges. Sensitivity shall not change by more than 1 mV/μT over the same temperature range.
(f) The zero shift due to residual magnetism shall be less than 3 mV after subjecting the magnetometer head assembly to a 2000 μT field.
(g) Frequency response shall be flat (that is, constant gain) from zero to 16.7 Hz.
(h) The sensitivity (gain) constant of the calibration winding shall be 6 microtesla per milliamperes (μT/mA) of calibration current, plus or minus 0.03 μT/mA.
(i) A 10 milliampere (mA) change in calibration current on one fluxgate shall not change the voltage output of either one of the other two fluxgates by more than 20 mV.
(j) Calibration windings shall not be adversely affected by calibration currents equal to or less than 25 mA.
(k) The excitation frequency shall be such that no matching of the fluxgates to the magnetometer circuits shall be necessary, if either needs to be changed.

3.4.5.8.1 Magnetometer accuracy. The actual magnitude of the magnetometer output voltage shall not differ from the ideal output voltage on an instantaneous basis by more than 0.06V for any one of or combination of the following conditions:

(a) Heading changes at any rate up to and including a maximum rate of 10 degrees per second.
(b) Roll rates up to and including a maximum rate of 30 degrees port to 30 degrees starboard in 3 seconds.
(c) Pitch rates up to and including a maximum rate of 5 degrees up to 5 degrees down in 1-1/2 seconds.
(d) Power source voltage and frequency variations within the tolerances specified in 3.4.4.1.
(e) Environmental variations within the tolerances specified in 3.16.
(f) Shock, vibration, and inclination conditions as specified in 3.11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

3.4.5.9 Degaussing power supplies. Degaussing power supplies shall be static, solid state, bipolar power supplies. Power supplies shall contain three operator adjustments, and shall be configured such that once adjusted, the power supply can be considered a black box by operating personnel. Power supplies shall receive power from the ship's service power distribution system, and signals from the magnetometer. Power supplies shall automatically supply controlled current to a degaussing loop and trouble signals to the control assembly.

3.4.5.9.1 Transfer function. Power supplies shall be of bipolar operation with the output current capable of being varied continuously from maximum positive to maximum negative, while passing smoothly through zero. The output currents shall be a function of the magnetometer output and operator settings as specified in the following formula:

$$I = (F_{T1}K_I + F_{T2})\ V_I \pm I_p$$

Where:

I = degaussing loop current in amperes (see 3.4.5.9.2).

F_{T1} and F_{T2} = operator settable transfer functions (see 3.4.5.9.4.1).

K_I = operator settable induced current gain (see 3.4.5.9.4).

V_I = magnetometer output voltage (see 3.4.5.8(b)).

$V_I = K\emptyset_M$ or $K\emptyset_A$ or $K\emptyset_L$ (depending on axis) where K = magnetometer gain (see 3.4.5.8(b)).

I_p = operator settable permanent current component (see 3.4.5.9.4).

3.4.5.9.2 Output. The rated output current shall be plus or minus 4.0 amperes d.c. into a degaussing loop with a nominal resistance of any value from 3.7 through 5.0 ohms, at loop cable temperature of 20°C.

3.4.5.9.2.1 Type of load. The degaussing loop will form a resistive and inductive load. The load inductance may be any value between zero and 100 millihenries (mH). Load resistance will vary up to 28 percent from nominal (see 3.4.5.9.2) with ambient temperature changes.

3.4.5.9.2.2 Output ripple current. The peak-to-peak ripple current shall not exceed 0.04 amperes for any load from "no load" to "full load", when measured with the equipment supplying a purely resistive load. Peak-to-peak values of any and all periodic and random deviations from nominal output current (including hum, noise, and spikes) shall be considered.

3.4.5.9.2.3 Self-limiting features. Power supplies shall limit the maximum obtainable output current to a value between 100 and 110 percent of rated output current. This feature shall prevent overloading equipment at extreme latitudes, with gain and permanent current components set for currents greater than 100 percent at lesser latitudes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

3.4.5.9.2.4 Output circuit isolation. The output circuit of each power supply shall be electrically isolated from the ship's service power supply.

3.4.5.9.2.5 Output circuit adjustments. Power supplies shall operate with load impedances of any value within the range specified (see 3.4.5.9.2 and 3.4.5.9.2.1), without adjustment or alinement of the power supply.

3.4.5.9.3 Accuracy. Actual magnitude of the power supply output current shall not differ from the ideal current (see 3.4.5.2) on an instantaneous basis by more than 0.04 amperes d.c. for any one of or combination of conditions specified in 3.4.5.8.1, and for changes up to 28 percent in load resistance due to temperature variations.

3.4.5.9.4 Operator adjustments. The induced current gain (K_I) shall be settable to any value between 0 and 0.667 amperes per volt (A/V). The permanent current component shall be settable to any value between -4.0 and +4.0 amperes. These controls shall be adjustable in increments not exceeding 0.5 percent of maximum setting.

3.4.5.9.4.1 An operator selectable and adjustable transfer function shall be provided to vary the phase response of the induced current component. The following transfer functions shall be provided:

(a) Case A: $F_{T1} = \frac{1}{1 + ST}$ and $F_{T2} = 0$

Where:
T is a variable to give a phase lag at 1 radian/sec of any angle between 0 and 45 degrees.
S is a Laplace operator.

(b) Case B: $F_{T1} = 1$ and $F_{T2} = ST$

Where:
T is a variable to give a phase lead of 90 degrees and a gain of 1 at 1 radian/sec.
S is a Laplace operator.

3.4.5.10 Test/calibration features. Test features, test circuits, and built-in meters shall be provided to:

(a) Measure the magnetometer outputs and the loop currents at the master cubicle without affecting the performance of the system in any way.
(b) Check, set, or reset current controls or subassembly gains and zero offset. Affecting system performance is permissible.
(c) Facilitate ranging/calibration of ship by providing undegaussed, normal degaussed, induced only and perm only signatures.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

3.4.5.10.1 A means shall be provided for the isolation of the degaussing loop from the power supply. When it is isolated, the degaussing loop shall be connected to two front panel monitor points to allow the degaussing loop resistance and insulation resistance to be checked.

3.4.5.10.2 A means shall be provided for the injection of time variant currents on one or more of the fluxgate calibration windings simultaneously.

3.4.5.11 Trouble/fault indicators. Fault circuits and fault indicators shall be provided to indicate trouble or fault if the degaussing system's performance or integrity is affected in any way. Troubles or faults indicated shall include:

(a) Excess errors.
(b) Failed or open current controls.
(c) Overheating conditions.
(d) Excessive ripple conditions.
(e) Blown fuses, or tripped circuit breakers.
(f) Open or shorted degaussing loops.
(g) Significant error in loop ampere-turns.
(h) Opens in interconecting cables between cubicles, magnetometer head, and bridge control unit.

3.4.5.11.1 Fault indicators and circuits shall be selective in the sense that they enable an operator to quickly isolate system malfunctions to a particular console, magnetometer, axis or power supply, and to a particular degaussing loop, or replaceable module.

3.4.5.11.2 A means shall be provided to automatically actuate alarms in the ship's control console in the event of a degaussing system failure, or in the event the master cubicle or any power supply cubicle is de-energized (due to power failure or turning the cubicle off). A set of electrically isolated, normally open, contacts, with a 6800-ohm, 5-watt, resistor connected across the contacts, shall be provided for each alarm circuit. The normally open contacts, when closed, will operate the remote alarm.

3.4.5.11.3 A warning indicator shall be provided to indicate if any test/calibrate switch is in other than the operate position.

3.4.6 Start-stop procedures. Special "turn-on" or "turn-off" procedures shall not be required. Equipment will normally be turned on and off by the power switch in the master cubicle. The equipment shall not be damaged and equipment operation shall not be affected by any possible means of turning equipment on or off or of changing current magnitudes or polarity.

3.4.7 Warm-up. The equipment shall operate normally immediately after being energized in any of the environments specified in 3.16. Immediately (a few seconds) after being energized, the degaussing coil currents shall be within 10 percent of the required currents. After a 1/2 hour warm-up period, the degaussing coil currents shall be within the values set prior to energizing (during last period of operation) and the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

equipment performance shall comply with the accuracy requirements specified in 3.4.5.8.1 and 3.4.5.9.3. Compliance with all of the requirements specified herein shall be obtained without making adjustments of any kind after energizing equipment or during warm-up period.

3.4.8 Magnetic signature. The equipment shall have a minimum magnetic signature. The equipment shall comply with the magnetic material, stray field, and eddy current requirements specified in MIL-E-16400.

3.4.9 Duty cycle. Equipment shall be capable of operating continuously with maximum rated output current to all loops simultaneously. This requirement is applicable with any possible combination of load resistances within the ranges specified.

3.4.10 Interconnection resistance. The equipment shall be such that no interconnecting cabling between units will require a maximum resistance value of less than 1.0 ohms per pair of conductors. This limitation shall not impair the system operation by sacrificing power gain, stability, or degree of accuracy specified herein. The purpose of this requirement is to minimize the size of interconnecting cables between units where the units of the system are installed remote from each other.

3.5 Electrical design. Electrical design requirements shall be in accordance with MIL-E-16400, and with the detail requirements specified herein.

3.5.1 Equipment protection. Protective devices shall be provided for protection of the equipment from damage due to faults in the equipment, grounds in the ship's service power supply, open or shorted degaussing coils, or primary power supply interruptions and transient variations as specified in DOD-STD-1399, section 300.

3.5.11 The equipment shall withstand, without damage, the tripping of circuit breakers, or the blowing of fuses, power source interruptions, transient variations, insulation resistance tests or ground detector tests described in DOD-STD-1399, section 300. After restoration of power, the equipment shall automatically perform its intended function without any adjustments. The manual resetting of breakers, and the like, shall not be required.

3.5.1.1.1 The equipment shall withstand, without damage, slowly varying, out of tolerance, power source voltages. The equipment shall not be damaged by power source voltage increase or decrease between zero volts and 130 percent of nominal voltage at any rate of change less than 1 volt per second. The manual resetting of breakers, and the like, is permissible for these conditions.

3.5.1.2 Fuse application. Circuits shall be designed and fuses applied such that random or unexplained blowing of fuses does not occur. The blowing of a fuse, due to causes other than part failure in equipment, grounds or out of specification variations in ship's service power supply, or degaussing loops shall be considered an equipment failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

3.5.1.3 Fail-safe features shall be included in the equipment design wherever feasible. Fail-safe operation is defined as the mode of operation where the degaussing loop currents are maintained at approximately their correct value or zero, when any part of the equipment fails. The equipment shall be designed such that fail-safe operation is obtained whenever any of the protective devices open or operate.

3.5.2 Electromagnetic interference. Equipment shall comply with the requirements of MIL-STD-461 for class A4 equipment. The equipment shall comply with the requirements for CE01, CE03, CS01, CS02, CS06, RE01, RE02, RS01, RS02, and RS03. The E-field level for the RS03 test shall be 50 volts per meter from 14 kilohertz (KHz) to 50 megahertz (MHz), and 10 volts per meter from 50 MHz to 10 gigahertz (GHz) for all equipment, (that is, magnetometer assembly, control unit and power supplies).

3.5.3 Electrical creepage and clearance distances. Minimum creepage and clearance distances between circuits and ground shall be in accordance with MIL-E-16400.

3.5.4 Indicator lights. Indicator light circuits shall be such that the same lamp (rating and style) is used for all applications.

3.6 Mechanical design. Mechanical design requirements shall be in accordance with MIL-E-16400.

3.7 Maintenance design.

3.7.1 Modular construction. The equipment shall use modular construction in accordance with MIL-E-16400.

3.7.1.1 Equipment shall be functionally divided into replaceable subassemblies in a manner that will make it feasible for technicians with limited experience, using simple readily available test equipment (see 3.7.2), to perform corrective maintenance on the equipment. Corrective maintenance on the equipment is defined as any and all maintenance required to restore the equipment to a satisfactory condition, except maintenance associated with the repair of the replaceable subassemblies.

3.7.1.2 "Plug-in" modular subassemblies shall be such that they can be interchanged between equipments without any mechanical or electrical adjustment, except magnetometer gain adjustments and current magnitude adjustments, and such that the system performance characteristics are identical to that obtained prior to the interchange of subassembly. "Plug-in" subassemblies shall be such that they cannot be inadvertently plugged into a wrong location.

3.7.1.3 Built-in test and monitoring systems shall make it possible to determine the satisfactory or unsatisfactory condition of each replaceable subassembly or module.

3.7.1.4 "Plug-in" modular subassemblies shall be replaceable without further degrading the degaussing system's performance during replacement. Subassemblies shall be replaced with the equipment on and operating. It shall not be necessary to turn the system, or any part of the equipment off to replace a subassembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

3.7.1.5 Accessibility. Accessibility of components shall be in accordance with requirement 36 of MIL-STD-454 and as specified herein. Unless otherwise specified (see 6.1.1), the equipment shall be constructed so that:

(a) Subassemblies can be removed without removing any other subassembly, printed circuit card or component.
(b) Chassis components can be removed without removing any other hardwired component, printed circuit card or subassembly.
(c) Adjustments can be made without removing any component, printed circuit card or subassembly.
(d) Subassemblies and printed circuit cards can be removed without the need to unsolder cables and interconnecting wiring (connections to all printed circuit cards and subassemblies shall be through pin and socket connectors).
(e) All cables from outside a cubicle shall connect to terminals accessible from the front of the cubicle. No access shall be required from the sides or rear for installation or maintenance purposes.

3.7.2 Test equipment. There shall be no need for any external test equipment to carry out routine monitoring to ensure correct operation of the equipment. Preventive maintenance shall be performed with the built-in test equipment.

3.7.3 Servicing. The equipment shall be fully operable with enclosures open for servicing. This requirement shall be met without the use of accessory test cables.

3.7.4 Internal controls and adjustments. The number of internal controls and adjustments shall be kept to a minimum consistent with the operation and maintenance of the equipment and the elimination of all need for the individual selection of parts. Accuracies and long term stability shall be constructed into the equipment by the judicious use of feedback circuits. All adjustments shall be independent of all other adjustments. The adjustement of one circuit element shall not affect the adjustment of any other circuit element.

3.7.4.1 Adjustable circuit elements for alinement and adjustment shall not be used unless specifically accepted by the contracting activity. Where adjustments are accepted, provisions shall be made to accomplish the required alinement. Adjustment controls shall be of a size, location, and operation to facilitate use.

3.8 Reliability. The contractor shall conduct the following reliability assurance tasks:

(a) Design reviews (see 3.8.2).
(b) Reliability prediction (see 3.8.3).
(c) Failure mode and effects analysis (see 3.8.4).
(d) Failure data collection, analysis, and corrective action (see 3.8.5).
(e) Reliability demonstration (see 4.8.18).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

3.8.1 The equipment mean time between failure (MTBF) shall be greater than or equal to 600 hours. The equipment is defined as a system with 72 power supplies in use (18 spare power supplies) and no redundancy.

3.8.2 Design reviews. The contractor shall establish and conduct a formal design review program. Participants shall include qualified contractor personnel from design, reliability, maintainability, and other pertinent areas of the contractor organization. NAVSEA representatives shall have the option to attend all formal design reviews. At least two formal design reviews shall be scheduled to allow for design changes if required, during the preliminary design stage and prior to final design. Typical items for discussion during the design review shall be:

(a) Reliability prediction status.
(b) Maintainability prediction status.
(c) Failure mode and effect analysis.
(d) Maintenance requirements.
(e) Operational and support concepts and requirements.
(f) Personnel constraints.
(g) Constraints associated with test equipment and special tools specified herein and in MIL-E-16400.
(h) Test points and test feature requirements associated with fault location and fault isolation specified herein and MIL-E-16400.

3.8.3 Reliability prediction. The contractor shall provide a reliability prediction report in accordance with the data ordering document included in the contract or purchase order (see 6.1.2). The report shall include the calculated MTBF for each equipment and for each modular subassembly. The reliability prediction shall be based on stress factors and part populations. Parts shall not be classified as critical or noncritical or by equipment operational mode.

3.8.4 Failure mode and effects analysis (FMEA). The contractor shall conduct a FMEA in accordance with MIL-STD-1629 and prepare a FMEA report in accordance with the data ordering document included in the contract or purchase order (see 6.1.2). The report shall provide input to the service manual. The analysis shall be based on a replaceable subassembly indenture level and a 60 day mission time.

3.8.5 Failure analysis and corrective action report. The contractor shall prepare a failure report in accordance with the data ordering document included in the contract or purchase order (see 6.1.2). The report shall include the following:

(a) Failure data collection, analysis, and corrective action shall include a closed loop failure data system for collecting, analyzing, and recording data on all equipment failures (fuse blowing and unscheduled alinements or adjustments shall be defined as failures and shall be included in the failure data system).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

(b) Data shall be recorded for failures discovered or occurring during:
(1) All phases of testing (in-plant or at Government facilities).
(2) Installation checkout of equipment by manufacturer's service engineers.
(3) Examination of equipment, subassemblies, or parts returned to manufacturer under warranty.

3.8.5.1 Failure. Failure is defined as occurring when the equipment ceases to function or its performance degrades below the requirements of this specification.

3.9 Maintainability. The contractor shall conduct the following maintainability assurance tasks:

(a) Design reviews (see 3.8.2).
(b) Maintainability prediction (see 3.9.2).

3.9.1 The equipment repair time (ERT) shall be less than or equal to 0.37 hours.

3.9.2 Maintainability prediction. The contractor shall predict maintainability parameter values for the equipment. The corrective maintenance geometric mean time to repair ($MTTR_G$) shall be predicted in accordance with part A, procedure II of MIL-HDBK-472, and the criterion specified in 3.9.2.1.

3.9.2.1 The equipment "method of repair" shall consist of replacing "plug-in" modular subassemblies or individual parts which are not located in modular subassemblies. "Plug-in" modular subassemblies shall be treated as "parts" for purposes of this prediction. Failure rates used for the maintainability prediction shall be the same as calculated for the reliability prediction (see 3.8.3). Maintenance task times shall be based on the criterion that all enclosures are closed and secured.

3.9.2.2 The contractor shall provide a maintainability prediction report in accordance with the data ordering document included in the contract or purchase order (see 6.1.2).

3.10 Life. The equipment shall have an expected life of 5 years (40,000 hours) of continuous operation without benefit of overhaul; and an expected life of 20 years with benefit of overhaul.

3.11 Shock, vibration, and inclination. Shock, vibration, and inclination shall be in accordance with MIL-E-16400. Inclination angle shall be 45 degrees. The upper frequency limit for vibration shall be 33 Hz.

3.12 Noise. The control equipment shall comply with the requirements of MIL-STD-740 for grade A, type 3 equipment, except that the airborne noise levels shall be less than the values specified in table I.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

TABLE I. Acceptable octave band sound pressure levels for machinery and equipment.

Decibels re: 20 micropascals (uPa)								
Octave band center frequency - Hz								
31.5	63	125	250	500	1000	2000	4000	8000
81	75	69	66	58	58	58	57	56

3.13 Safety. Safety requirements shall be in accordance with MIL-E-16400.

3.14 Controls, indicators, and panel layouts. Controls, indicators, and panel layouts shall be in accordance with MIL-E-16400.

3.15 Processes. All processes such as soldering, welding, corrosion-resisting treatments, painting, and encapsulation shall be in accordance with the requirements of MIL-E-16400.

3.16 Environmental conditions. Environmental requirements shall be in accordance with MIL-E-16400 for service conditions as follows:

(a) Magnetometer head assembly - exposed-unsheltered (ship), temperature range 2.
(b) All other equipment - sheltered-controlled (ship), temperature range 4.

3.17 Parts. The requirement for parts shall be in accordance with MIL-E-16400 and the additional requirements specified herein.

3.17.1 Electron tubes shall not be used.

3.17.2 Nonstandard parts. The contractor shall prepare a nonstandard parts documentation and certificate of completion in accordance with the data ordering document included in the contract or purchase order (see 6.1.2), and as specified in 3.17.2.1 and 3.17.2.2.

3.17.2.1 The number of nonstandard parts used shall be the minimum number necessary to meet the requirements for the equipment. The use of two different nonstandard parts will not be accepted if one of the two can be used for both applications.

3.17.2.2 Matched parts. If matched parts are required for proper performance of the equipment, the matched parts shall be treated as nonstandard parts and acceptance obtained prior to their use in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

equipment. Sets of matched parts shall be identified by a unique part number and packaged in a manner such that the set can be removed, acquired, and replaced as a subassembly.

3.17.2.3 Nonrepairable subassemblies. Nonrepairable subassemblies shall be considered nonstandard parts.

3.18 Designation and marking.

3.18.1 Identification plates. Identification plates and identification plate markings shall conform to type A, B, or C of MIL-P-15024 and MIL-P-15024/5. The contractor shall obtain the applicable stock number from the contracting activity for inclusion on the identification plate prior to shipping. Serial numbers shall be assigned in accordance with MIL-E-16400.

3.18.2 Instruction plates. Operating instructions shall be in accordance with MIL-E-16400 with respect to requirements for lettering, legibility, materials, and the like.

3.19 Drawings. The contractor shall prepare drawings in accordance with the data ordering document included in the contract or purchase order (see 6.1.2), and as specified in 3.19.1 through 3.19.4.1.

3.19.1 Engineering drawing types shall include detail parts drawings, assembly drawings, control drawings, installation drawings, diagrammatic drawings, special purpose drawings, and parts lists. Contractor design activity identifiers and drawing numbers shall be assigned to all engineering drawings and lists. Mono-detail drawings shall not be required. Specification control drawings or source control drawings shall be prepared for all vendor or commercial items that are not covered by Government or nationally recognized industry association standards or specifications. Parts lists shall be prepared for all assembly and subassembly drawings. The parts lists may be integral with or separate from the drawings.

3.19.2 Design evaluation drawings. Design evaluation drawings shall consist of assembly drawings, parts and material lists, schematic diagrams, block diagrams, wiring diagrams, maintenance information (see 3.19.4), and other drawings as required to provide information necessary to evaluate the design, including the maintenance concept, support concept and interfaces.

3.19.2.1 Schematic diagrams, wiring diagrams, and block diagrams shall be drawn in a manner such that they will be suitable for inclusion as illustrations in the manual (see 3.20).

3.19.3 Test procedures (see 4.5) shall be assigned drawing numbers and referenced on the assembly drawings for the equipment. Test procedures shall be considered drawings and supplied with engineering drawings. Test procedures shall be referenced on assembly drawings by "test per . . . ." drawing notes.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

3.19.4 Maintenance information. Equipment maintenance information submitted with the design evaluation drawings shall include the following:

(a) Functional description of the overall equipment, of each functional section and subsection, of each modular assembly, and of each circuit or stage in functional sections or modular assemblies.
(b) Overall block diagrams, functional section block diagrams and simplified schematic diagrams for equipment, for each functional section, and for each circuit stage (or portions thereof).
(c) Adjustment, alinement and repair procedures.

3.19.4.1 The maintenance information submitted need not be prepared in a manner such that it is suitable for inclusion in the equipment manual; however, the form and content of this information shall be in accordance with the requirements for the manual. Functional descriptions, adjustment procedures, and simplified schematics shall be identified by reference designations and functional names such that they can be related to each other and to block diagrams and maintenance schematic diagrams submitted as drawings. The information need not be assembled in any particular order and the simplified schematics may be legible sketches.

3.20 Equipment manual. The contractor shall prepare a manual in accordance with the data ordering document included in the contract or purchase order (see 6.1.2) and as specified in 3.20.1 through 3.20.3.2.

3.20.1 Distribution statement. The distribution statement to be included on the title page shall be as follows:

"This document is subject to special export controls and each transmittal to foreign governments or foreign nationals may be made only with prior approval of the Naval Sea Systems Command."

3.20.2 The General Information section of the manual shall include information explaining the basic principles of and the need for degaussing. This information shall be in addition to the overall description of the functions and purpose of the equipment.

3.20.3 Troubleshooting diagrams. Troubleshooting diagrams shall consist of a power distribution diagram, maintenance schematic diagrams, intra-unit interconnection diagrams and troubleshooting-maintenance dependency-matrix charts.

3.20.3.1 Maintenance schematic diagrams shall be prepared so as to illustrate the functions of specific circuit arrangements and to facilitate tracing the circuit and its functions without regard to the actual physical size, shape, or location of assemblies or parts. Maintenance schematic diagrams shall be drawn to illustrate major functional sections. Individual modules or printed circuit boards shall not be illustrated by separate schematic diagrams.

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:19Z
Check the source to verify that this is the current version before use.

3.20.3.2 Intra-unit interconnection diagrams, or wire lists, shall be prepared to illustrate the physical connections of all assemblies, subassemblies, and parts in such detail as needed to make or trace any and all connections in the equipment.

3.21 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time (see 6.1.2). Verification will consist of (a) surveillance of the operations to determine that practice, methods, and procedures of the written system description are being properly applied (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all the requirements of this specification. Failure of the contractor to correct promptly deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality program. The contractor shall provide and maintain a quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with the data ordering document included in the contract or purchase order (see 6.1.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.6).
(b) Quality conformance inspection (see 4.7).

4.3 Test conditions. Except for those tests where the following factors are variables (or otherwise specified herein), tests shall be conducted with the equipment operating under the following conditions:

(a) Ambient temperature - as specified in MIL-E-16400.
(b) Relative humidity - as specified in MIL-E-16400.
(c) Atmospheric pressure - as specified in MIL-E-16400.
(d) Supply voltage and frequency - nominal.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

4.3.1 Performance tests as specified in 4.8.3.1.2 and 4.8.6 shall be conducted with the magnetometer head located in a known magnetic field. This field shall be a magnetic field generated by a Helmholtz or Ruebens coil. The magnetic field value shall be measured by a standard instrument having sufficient accuracy to ensure that the equipment under test is in compliance with all the requirements of this specification. Data on the proposed standard instrument's accuracy and substantiation of its calibration traceable to the National Bureau of Standards, or an equivalent testing station, satisfactory to NAVSEA shall be submitted to NAVSEA for review (see 6.1.2)

4.4 Adjustments, alinement or repairs. The equipment shall be completely debugged, and all features shall be operating in accordance with the requirements of this specification before starting Government witnessed tests. After testing is in progress, adjustments, alinements, or repairs (including the replacement of fuses and adjustment of current controls) will be permitted only under the following conditions:

(a) Take and record (in test report (see 4.6.2)), required measurements before making adjustments and repairs.
(b) Notify and obtain Defense Contract Administration Services Management Area (DCASMA) acceptance before making adjustments or repairs.
(c) Record adjustments or repairs (or fuses replaced) made and why the adjustments or repairs were necessary in the test report.
(d) Repeat any testing already completed that could possibly be affected by the adjustment or repair.

4.4.1 Adjustments and alinements made during the "Operating Test" (see 4.8.2), shall not be subject to the conditions specified in 4.4.

4.5 Test procedures. The contractor shall prepare test procedures in accordance with the data ordering documents included in the contract or order (see 6.1.2) and as specified in 4.5.1 and 4.5.2:

4.5.1 First article inspection procedures. First article inspection procedures shall be prepared for all first article tests that are not covered by quality conformance test procedures. Quality conformance test procedures shall be used for all first article tests that are identical to quality conformance tests.

4.5.2 Quality conformance inspection procedures. Quality conformance inspection procedures shall be prepared for each item of equipment and each modular subassembly offered for delivery. These procedures shall provide all required quality conformance inspection and acceptance criteria for the item(s) covered. These procedures shall be used for the inspection of complete shipsets of equipment and for the inspection of individual magnetometer head assemblies, and modular subassemblies acquired as spares for logistic support of equipment.

4.6 First article inspection. First article inspection shall include examination and tests specified in table II and shall be performed on the first equipment(s) as specified (see 6.1.1).

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:19Z
Check the source to verify that this is the current version before use.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Examination and test paragraph
Surface examination:		
Workmanship	3.21	4.8.1(a)
Materials and parts	3.3 & 3.17	4.8.1(b)
Finishes	3.15	4.8.1(c)
Safety	3.13	4.8.1(d)
Designations and markings	3.18	4.8.1(e)
Creepage and clearance distances	3.5.3	4.8.1(f)
Operating	3.4.5	4.8.2
Magnetometer design	3.4.5.8	4.8.4
Weights and dimensions	3.4.5.1.1	4.8.5
System accuracy	3.4.5.8.1	4.8.7
Phase lag	3.4.5.9	4.8.8
Ripple	3.4.5.9.2.2	4.8.12
Interchangeability	3.7.1.2	4.8.13
Warm-up time	3.4.7	4.8.9
Performance	3.4.5	4.8.3
Accuracy versus supply variations	3.4.5.8.1 & 3.4.5.9.3	4.8.11
Shock	3.11	4.8.16
Vibration	3.11	4.8.17
Inclination	3.11	MIL-E-16400
Harmonic current	MIL-E-16400	MIL-E-16400
Power	3.4.4.1	4.8.6
Dielectric withstanding voltage, insulation resistance	MIL-E-16400	MIL-E-16400
Leakage current	MIL-E-16400	MIL-E-16400
Equipment protection	3.5.1	4.8.14
Temperature and humidity	3.16	MIL-E-16400
Sunshine	MIL-E-16400	MIL-E-16400
Enclosure	MIL-E-16400	MIL-E-16400
Thermal design	MIL-E-16400	MIL-E-16400
Electromagnetic interference	3.5.2	4.8.15
Noise	3.12	MIL-E-16400
Accelerated life (reliability)	3.8	4.8.18

4.6.1 Order of tests. First article inspection shall be conducted in any order or sequence convenient to the testing activity, except as required for compliance with the following:

(a) Surface examination, operating, performance, and ripple tests shall be conducted before all other tests.
(b) The vibration test shall be conducted after the shock test to aid in the discovery of loose or weakened parts.
(c) Accelerated life test shall be conducted after all other tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

4.7 Quality conformance inspection. Quality conformance inspection shall be conducted on each shipset of equipment, and on each individual equipment item, including spare parts, modules, and assemblies offered for delivery. Equipment shall be subjected to the examination and tests specified in table III.

TABLE III. Quality conformance inspection.

Inspection	Requirement paragraph	Examination and test paragraph
Surface examination:		
Workmanship	3.21	4.8.1(a)
Materials and parts	3.3 & 3.17	4.8.1(b)
Finishes	3.15	4.8.1(c)
Safety	3.13	4.8.1(d)
Designations and markings	3.18	4.8.1(e)
Creepage and clearance distances	3.5.3	4.8.1(f)
Insulation resistance	MIL-E-16400	MIL-E-16400
Dielectric withstanding voltage	MIL-E-16400	MIL-E-16400
Operating	3.4.5	4.8.2
Performance	3.4.5	4.8.3

4.7.1 Quality conformance inspection procedures. Quality conformance inspection shall be conducted in accordance with the detailed inspection procedures used and accepted for first article inspection.

4.8 Test methods.

4.8.1 Surface examination. Equipment shall be given a thorough examination to determine conformance to the applicable requirements and accepted drawings for the following:

(a) Workmanship, assembly and fit (see 3.21).
(b) Materials, parts (see 3.3 and 3.17).
(c) Finishes (see 3.15).
(d) Safety requirements (see 3.13).
(e) Designations and markings (see 3.18).
(f) Creepage and clearance distances (see 3.5.3).

4.8.2 Operating test. The equipment shall be energized and subjected to an operating test to insure qualitatively the proper functioning of the equipment, including all operating controls, test features, test controls, adjustments, indicating devices, trouble circuits, and safety features such as interlocks and discharge circuits.

4.8.2.1 The operating test procedures shall include procedures for adjusting all adjustable circuit elements (such as potentiometers, resistors, capacitors, and inductors, etc.) in magnetometer, control unit, power supplies, and modular subassemblies. If selected parts (resistors, capacitors, etc.) are used, the operating test procedures shall include procedures for selecting these parts. Adjustable circuit elements shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

adjusted, and all selected parts shall be selected, during the operating tests. These devices shall not be readjusted or reselected during performance or environmental tests that follow operating tests (NOTE: This requirement does not apply to the current adjust controls or test controls.)

4.8.2.2 If it is necessary to adjust a circuit element to demonstrate a feature (i.e., trouble circuit operation), then the procedure shall be prepared to combine the demonstration with the adjustment, with final adjustment as the last step.

4.8.3 Performance tests.

4.8.3.1 Magnetometer performance tests. The complete magnetometer (that is, head assembly, mounting brackets, magnetometer cable, connectors, electronic circuits in the control assembly, etc.) shall be tested in accordance with 4.8.3.1.1 through 4.8.3.1.3.

4.8.3.1.1 Orthogonality of sensing elements. The magnetometer head assembly, mounted on a test turntable by means of an orthogonal block, shall be rotated in the earth's magnetic field about each of its three axes to determine compliance with 3.4.5.3. The test shall be conducted as follows:

(a) The orthogonal block shall be mounted on the test turntable, with the block oriented with respect to the turntable axis of rotation such that the magnetometer head assembly mounted on the orthogonal block will be rotated about an axis parallel to the axis of one of the fluxgates and perpendicular to the axis of the other two fluxgates.
(b) The test turntable shall be located in a magnetically clean (that is, gradient free) area. The test turntable shall be oriented such that the axis of rotation is parallel to the axis (direction) of the earth's magnetic field. The test turntable shall be locked in this position such that the orientation of the axis rotation of the turntable does not vary during the entire test.
(c) The magnetometer shall be mounted on the orthogonal block such that the axis of the vertical (M coil) fluxgate is parallel to the axis of rotation and the axis of the longitudinal and athwartships fluxgates are perpendicular to the axis of rotation.
(d) The magnetometer shall be rotated 360 degrees and the output voltages from all three fluxgates shall be measured simultaneously every 30 degrees. The measured voltages and angles shall be recorded. Each set of data shall consist of 12 angles and 36 voltages.
(e) The magnetometer shall be remounted on the orthogonal block such that the axis of the longitudinal (L coil) fluxgate is parallel to the axis of rotation. The magnetometer shall be rotated as specified in (d) herein and a second set of data recorded.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

(f) The magnetometer shall be remounted on the orthogonal block such that the axis of the athwartship (A coil) fluxgate is parallel to the axis of rotation. The magnetometer shall be rotated as specified in (d) herein and a third set of data recorded.

(g) The three sets of data shall be used to determine orthogonality of the fluxgates and alinement of the fluxgates with respect to the mounting surface and azimuth pins (see 3.4.5.3).

4.8.3.1.2 Static accuracy. The magnetometer head assembly shall be located in known magnetic fields (see 4.3.1). The magnetic field along each axis shall be varied from minus 60 μT to plus 60 μT. The three magnetometer output voltages shall be measured and recorded with field set of plus or minus 60, plus or minus 30, and 0 μT (5 points) to determine compliance with 3.4.5.8.

4.8.3.1.3 Calibration winding accuracy. The magnetometer head assembly shall be located in a constant magnetic field (earth's field), and the magnetic field along one fluxgate shall be varied by varying the current in the calibration winding for the fluxgate. The calibration current shall be adjusted for plus 6.0 volts output from the fluxgate, and the three output voltages (that is, output of each fluxgate), and the calibration current recorded. Calibration current shall then be readjusted for minus 6.0 volts from the fluxgate and the three output voltages and current recorded. This test shall then be repeated for the other two fluxgates. The data (6 currents, and 18 voltages) shall be used to determine compliance with 3.4.5.8.

4.8.3.2 Power supply performance tests. Each power supply shall be tested to demonstrate that the power supply complies with requirements specified in 3.4.5.9.

4.8.4 Magnetometer design. The magnetometer shall be set up in accordance with 4.8.3.1 and tested in accordance with 4.8.4.1 through 4.8.4.3 to determine compliance with 3.4.5.8.

4.8.4.1 Residual magnetism. The magnetometer head assembly shall be placed in a zero μT magnetic field and the output voltages measured and recorded. The magnetometer head assembly shall then be exposed to a positive magnetic field of at least 2000 μT for at least 10 seconds. The direction of the 2000 μT field shall be along the axis of the connector that connects the magnetometer head to the magnetometer cable. The 2000 μT field shall be removed and the magnetometer output voltages in a zero μT field shall again be measured and recorded. The magnetometer head assembly shall then be exposed to a negative 2000 μT field for at least 10 seconds. The negative field shall be removed and the magnetometer output voltages in a zero μT field shall again be measured and recorded.

4.8.4.2 Temperature drift. The magnetometer head assembly shall be located in a magnetically clean, gradient free, area in a nonmagnetic temperature chamber. The magnetometer shall be mounted such that the orientation of the sensing elements with respect to the earth's magnetic

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

field does not change during the test. The chamber temperature shall be varied such that the internal temperature of the magnetometer head is lowered from room ambient (about 20°C) to minus 28°C, then raised from minus 28°C to plus 65°C, and then lowered from plus 65°C to ambient. The magnetometer output voltages shall be monitored and recorded at temperature extremes and periodically while temperature is changing. The magnetometer sensitivity (gain) shall be measured at ambient and extreme temperatures (see 4.8.4.2.2).

4.8.4.2.1 The internal temperature of the magnetometer head shall be monitored with a thermocouple installed inside the case or by measuring the change in resistance of one of the windings with a Kelvin bridge (or Wheatstone bridge if winding resistance is large enough). The internal temperature need not be stabilized at extreme temperatures. The low temperature sensitivity can be measured with an internal temperature between minus 23°C and minus 28°C. The high temperature sensitivity can be measured with an internal temperature between plus 60°C and plus 65°C.

4.8.4.2.2 The sensitivity (gain) of each axis shall be measured by recording the calibration currents required to obtain output voltages of plus 10 volts, zero volts, and minus 10 volts.

4.8.5 Weights and dimensions. The weights and the dimensions (overall and mounting) of all equipment comprising the complete equipment shall be determined, recorded, and compared with the applicable requirements and values given on the accepted drawings and in the manual.

4.8.6 Power. The input voltages, currents and wattages, and the output voltages and currents shall be measured and recorded for minimum, maximum, and one-half load conditions. After the data is taken, the percent load unbalance shall be calculated in accordance with DOD-STD-1399, section 300, to determine compliance with 3.4.4.1.

4.8.7 System accuracy. The equipment shall be set up and tested to demonstrate that the equipment complies with the accuracy requirements specified in 3.4.5.8 and 3.4.5.9. The magnetometer shall be located in a known magnetic field (see 4.3.1) and the magnetic field along each of the three magnetometer axes shall be varied from minus 60 µT to plus 60 µT in 10 µT steps. Output currents shall be measured with the meter connected in series with the power supply load. The applied field, the output currents, and the currents and fields indicated by built-in test features shall be recorded to demonstrate compliance with requirements.

4.8.7.1 Dynamic accuracy. The equipment shall be set up and tested to demonstrate dynamic accuracy. A dynamic field (0 through plus or minus 30 µT) shall be impressed on the magnetometer fluxgates and field, outputs, and errors monitored with an oscillograph. Data shall be taken for each of the three magnetometer axes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

4.8.7.1.1 Provisions shall be made to record on an oscillograph the applied field, the indicated field, the power supply output current, the magnetometer error (difference between normalized field and normalized magnetometer output voltage), and the power supply error (difference between normalized input and output). Recordings of input, outputs and errors shall be made simultaneously for the following conditions:

(a) Static response to obtain data to validate calibration of error and output traces. Magnetometer and power supply errors of one percent shall be obtained by changing gains. The recorder shall be run for a few seconds while field, magnetometer output voltage, and power supply current are measured with accurate meters. Measured values shall be written on the oscillograph.
(b) Dynamic field frequency of 1/6 Hz.
(c) Dynamic field frequency as required to obtain one percent error from either the magnetometer or power supply.

4.8.7.2 Oscillographs shall be given an identifying number and annotated as required to record all significant information, such as; date, test operator, equipment, oscillograph identification, scale factors, amplitudes, and frequencies.

4.8.8 Phase lag. Equipment shall be set up in accordance with 4.8.7.1, and provisions made to record applied field and output current simultaneously on oscillographs. Recordings shall be calibrated such that compliance with requirements of 3.4.5.9 for phase shift and attenuation can be determined. Recording shall be made with signal rate of 1/6 Hz, and with phase lag control set for 0, 15, 30, and 45 degrees phase lag. One power supply per axis per power supply cubicle shall be converted to an eddy current only configuration (see 3.4.5.9.4.1, case B) and the phase angle checked for 90 degree lead. Oscillographs shall be numbered and annotated in accordance with 4.8.7.2, and actual phase lag and attenuation calculated and recorded in the test report.

4.8.9 Warm-up time. The master cubicle and the magnetometer head assembly shall be simultaneously placed in ambient temperatures of minus

$3 \pm 5^{\circ}C$ for a period of not less than 8 hours with the equipment de-energized.

4.8.9.1 The equipment shall then be energized and the operation shall be monitored immediately to determine compliance with requirements specified in 3.4.7. After the equipment has been energized for 1/2 hour period, outputs of equipment shall be monitored to determine compliance with requirements specified in 3.4.7.

4.8.10 Reference measurements. Equipment shall be set up and tested in accordance with 4.8.10.1 prior to, during, and after environmental tests to determine compliance with environmental requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:19Z
Check the source to verify that this is the current version before use.

4.8.10.1 Meters shall be connected to measure magnetometer output voltage (V_m), power supply output currents (I_L) and calibration winding current (I_c). Output currents from three power supplies per cubicle shall be recorded. Permanent current components for the power supplies shall be set for 0, 50 percent, and 100 percent of maximum rated permanent current. Procedure for taking one set of data for one axis shall be as follows:

(a) Adjust I_c for zero milliamperes. Record V_m and I_L.

(b) Adjust I_c for V_m = plus 6.0 volts. Record I_c and I_L.

(c) Adjust I_c for V_m = zero volts. Record I_c and I_L.

(d) Adjust I_c for V_m = minus 6.0 volts. Record I_c and I_L.

4.8.11 Accuracy versus supply variations. Equipment shall be tested to determine effects of variations in voltage and frequency on accuracy. Performance tests shall be conducted for each of the steady-state voltage and frequency conditions specified in MIL-E-16400. The equipment shall be set up and tested in accordance with 4.8.10. The magnetometer shall be located or shielded such that the ambient field at the magnetometer does not vary during tests. Data shall be taken for at least two axes.

4.8.12 Ripple. The equipment shall be tested to determine compliance with requirements of 3.4.5.9.2.2. Measured ripple voltage on power supply outputs shall be recorded for worst case conditions (for example: supply voltage, load resistance, and output currents set for maximum ripple; all power supplies set at maximum output simultaneously). Enough data shall be recorded to indicate worst case conditions.

4.8.13 Interchangeability test. The interchangeability test shall be performed to determine electrical and mechanical interchangeability. The equipment shall be set up and tested in accordance with 4.8.10. Reference measurements shall be taken for all three axes. The modular "plug-in" subassemblies shall be interchanged at random. The magnetometer head assembly shall be interchanged with a spare magnetometer head assembly. Reference measurements for all three axes shall be taken again and examined to determine compliance with 3.7.1.2.

4.8.14 Equipment protection. Equipment shall be tested to determine conformance with 3.5.1 through 3.5.1.1.1 when subjected to the following conditions:

(a) Open, shorted and grounded load resistances.
(b) Supply voltages of any value between zero and 130 percent of the rated supply voltage and any time duration.
(c) Power interruption test in accordance with MIL-E-16400.
(d) Transient voltage test in accordance with MIL-E-16400.
(e) Transient frequency test in accordance with MIL-E-16400.
(f) Spike voltage test in accordance with MIL-E-16400.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

The equipment shall be tested after being subjected to each of the above conditions as appropriate. After the equipment has been subjected to all of the above conditions, reference measurements in accordance with 4.8.10 shall be taken and compared to reference measurements taken prior to equipment protection tests.

4.8.14.1 Equipment shall be tested to determine conformance with start-stop procedures specified in 3.4.6.

4.8.15 Electromagnetic interference. The equipment shall be tested in accordance with MIL-STD-462 to demonstrate compliance with 3.5.2 for both conducted and radiated emissions and susceptibility of equipment.

4.8.16 High impact (HI) shock test. The equipment shall be shock tested in accordance with MIL-S-901 for grade A, type A, class I equipment. Requirements for test shall be as follows:

(a) Mounting. The equipment shall be mounted on the shock machine in a manner simulating shipboard installation in accordance with MIL-S-901. Equipment which is complete in more than one enclosure shall be tested complete by shock testing all assemblies.

(b) Mode of equipment operation during tests. The equipment shall be set up, energized and operated during the shock test. An output current and the indicated field strength of the L axis shall be monitored with oscillograph immediately before and after and during each shock below. Other outputs shall be monitored as appropriate. After shock test, equipment operation shall be completely tested to determine equipment was not damaged and reference measurements in accordance with 4.8.10 for all three axes shall be taken and recorded to determine that equipment alinement and adjustments were not affected by the shock test.

(c) A definition of failure to perform principal functions:

(1) Breakage of any parts including mounting bolts. Minor chipping of parts such as plastic knobs and cases and minor distortion of parts will be permitted where such chipping or distortion cannot in any manner impair operation of the equipment as specified.

(2) Appreciable distortion of any parts, including enclosure and framework.

(3) A value of insulation resistance lower than that permitted by this specification.

(4) Failure to pass surface examination. The equipment shall be carefully examined after removing all removable panels and doors to ascertain any mechanical damage. If requested by the DCASMA, partial disassembly shall be performed to aid in determining any possible damage.

(5) Failure to perform any electrical tests requested by the Government during and following the shock test. No adjustment or replacement of damaged parts shall be permitted during shock test, unless specifically permitted by NAVSEA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

(d) Disposal of shock tested equipment. Equipment which has been subjected to HI shock tests shall be accepted as production equipment of the contract or order only under the following conditions:

(1) The equipment meets all terms and conditions of the contract for acceptance and is restored to a like new condition.

(2) Not more than one shock tested equipment per contract or order is being offered for acceptance following HI shock test.

(3) An identification plate is added identifying the unit as HI shock tested.

4.8.17 Vibration test. The equipment shall be subjected to type I vibration test in accordance with MIL-STD-167-1. Equipment operation during test, shall be the same as specified for shock tests (see 4.8.16), except that an oscillograph need not be used to monitor outputs.

4.8.17.1 After completion of the shock and vibration tests, the magnetometer head assembly shall be tested in accordance with 4.8.3.1.1 to demonstrate that the alinement of the magnetometer fluxgates was not affected.

4.8.18 Accelerated life test. The equipment shall be subjected to accelerated life tests specified in MIL-E-16400, except the test period shall be 500 hours in lieu of 360 hours. The test period shall be increased to 500 hours by repeating the high temperature voltage/frequency cycling tests at 95 percent relative humidity for 94 cycles instead of 59 cycles.

4.8.18.1 During accelerated life tests, equipment shall be set up and operated as follows:

(a) The magnetometer head assembly shall be oriented such that the M axis output is about plus 40 to plus 60 uT and the A and L axes outputs have a positive polarity and about the same magnitude.

(b) The A and L axes calibrate winding shall be connected to one low frequency function generator. The jacks shall be connected such that the same calibration current, generated by the function generator, flows through the calibration winding on the A fluxgate and the L fluxgate.

(c) The current controls on some A and L coil power supplies shall be set for a lagging current, some for leading current, and some for in-phase currents. Some power supplies shall be set for zero permanent current component, some for maximum positive perm, and some for maximum negative perm.

(d) The function generator shall be adjusted to produce a calibration current changing at a rate of 1/6 Hz with a magnitude that causes the A and L currents to vary from a positive limiting condition to a negative current, that is, the current polarity shall reverse during each half cycle.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

4.8.18.2 During accelerated life tests, the performance shall be checked at the specified intervals by turning the function generator off and performing reference measurements on all three axes, in accordance with 4.8.10. After completion of the accelerated life test, the equipment operation shall be completely tested to determine equipment was not damaged and equipment alinement and adjustments were not affected.

4.8.18.3 A malfunction causing "shutdown", the operation of a protective device (see 3.5.1), or "out of specified limit" performance during the accelerated life test shall be considered as a failure requiring corrective action. Upon completion of corrective action, the test shall be restarted and run according to the following conditions, until a satisfactory 500-hour run is completed.

(a) If failure occurs during the first 25 percent of the endurance test, restart the test from "0" time and continue test to complete the 500-hour test.
(b) If failure occurs between 25 and 100 percent of the endurance test, restart from time equal to hours when failure occurred minus 125 and continue test to complete the 500-hour test.
(c) A repeat failure of the same part, similar parts performing the same function, or similar parts performing dissimilar functions shall be cause to discontinue the test to correct the deficiency. If the test is discontinued, action shall be taken to determine the cause of failure and to correct the deficiency. Verification of the corrective action shall be accomplished by a continuation of test, wherein the test item shall accumulate at least 250 hours of test time without recurrence of the failure.

4.8.18.4 The equipment shall be accepted upon successful completion of the 500-hour test with less than three failures. In the event that more than two failures occur, the test shall be discontinued, and NAVSEA advised of failures and proposed corrective action. NAVSEA shall be provided with copies of all accelerated life test records and an updated failure report (see 3.8.5). NAVSEA will review information provided and supply conditions for repeating test.

4.9 Packaging inspection. Preservation-packaging, packing, and marking shall be inspected to determine conformance with the requirements of section 5 of this specification.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 General requirements.

5.1.1 Preservation-packaging, packing, and marking data. The contractor shall prepare packaging and transportation data, drawing(s), and report(s) of the preservation-packaging and packing test(s) in accordance

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

with the data ordering document specified in the contract or purchase order (see 6.1.2). The data shall be submitted to the acquisition contracting office (ACO), via the technical packaging element of the ACO.

5.1.2 Previous submittal. Technical data submitted as specified in 5.1.1 for evaluation on previously acquired identical equipment and parts is not required.

5.1.3 First article sample. When level A preservation-packaging and level A or B packing are required by the contract or order, a first article pack of one equipment and its shipping container(s) shall be tested as specified in MIL-E-17555 prior to beginning packaging production, unless otherwise specified (see 6.1.1).

5.1.3.1 Dummy load. Unless otherwise specified (see 6.1.1), a dummy or simulated load shall be used for the rough handling tests in accordance with appendix C of MIL-STD-794.

5.2 Preservation-packaging, packing, and marking. Each automatic degaussing and control equipment shall be preserved-packaged level A or C, packed level A, B or C, and marked in accordance with MIL-E-17555 as specified (see 6.1.1).

5.2.1 Technical data (see 3.22). Drawings, manuals, miscellaneous reports, etc. to be forwarded the contracting agency in compliance with the requirements of this specification (see 6.1.2), shall be packaged and packed level C in accordance with the applicable data specification. Where no packaging requirements exist, data shall be prepared for delivery in accordance with the level C requirements of MIL-E-17555.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether first article sample and tests are required (see 3.1 and 4.6).
(c) Number of sets of alinement tools required (see 3.4.5.5.1).
(d) If construction is other than specified (see 3.7.1.5).
(e) When first article preservation-packaging and packing test(s) are not required (see 5.1.3).
(f) When a dummy load is not required for testing (see 5.1.3.1).
(g) Levels of preservation-packaging and packing required (see 5.2).

6.1.2 When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:19Z
Check the source to verify that this is the current version before use.

Paragraph	Data requirement	Applicable DID	Option
3.8.3	Reliability predictions report	DI-R-7082	
3.8.4	Failure modes and effects analysis report	DI-R-30507	
3.8.5	Report, failure/malfunction	UDI-T-23724	
3.9.2.2	Reports, maintainability analysis and prediction	DI-R-2128	
3.17.2	Nonstandard part approval requests/proposed additions to an approved program parts selection list	DI-E-7028	
3.19	Drawings, engineering and associated lists	DI-E-7031	Level 3
3.20	Manual, technical, preliminary	DI-M-2043	
3.20	Technical manual, manuscript copy	DI-M-2042	
4.1.1	Notification of tests	DI-T-23731	
4.1.2	Quality assurance program plan requirements	DI-R-4802	
4.3.1	Procedures, instrument calibration (ICP)	UDI-T-22703	
4.5	Procedures, test	UDI-T-23732	
4.5	First article inspection procedures	DI-T-4901	
4.5	First article inspection report	DI-T-4902	
5.1.1	Packaging and transportation support data	UDI-P-23508	
5.1.1	Preservation-packaging and packing test report	UDI-T-23766	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.1.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

6.2 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:19Z
Check the source to verify that this is the current version before use.

6.3 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.3.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 1075-N041)

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-E-22437E(SH)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:19Z
Check the source to verify that this is the current version before use.